

Emanuele Bugliarello

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Research Interests

Multimodal and multilingual natural language understanding, representation learning, transfer learning

Education

PhD in Computer Science
University of Copenhagen
Advisors: Desmond Elliott, Anders Søgaard
2019–present (Jan 2023)
Copenhagen, Denmark

MSc in Communication Systems, Data Analytics specialisation
École Polytechnique Fédérale de Lausanne (EPFL)
Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)
2015–2018
Lausanne, Switzerland

BEng in Electronics and Information Engineering
Tongji University (同济大学)
Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30
2013–2018
Shanghai, China

BSc in Telecommunications Engineering
Politecnico di Torino
GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 *Cum Laude*
2012–2015
Turin, Italy

Machine Learning Schools: LxMLS 2020, Oxml 2020, ALPS 2021

Experience

Visiting Scholar with Prof. Siva Reddy
Mila
Upcoming (Winter 2021)
Montreal, Canada

Research Scientist Intern with Rishabh Mehrotra and Mounia Lalmas
Spotify Research
On Multi-Objective Music Sequencing via Counterfactual Beam Search
08/2020–11/2020
London, UK

Research Associate with Prof. Naoaki Okazaki
Okazaki Lab @ Tokyo Institute of Technology (東京工業大学)
On Neural Machine Translation: Dependency-aware Self-Attention and Cross-Mutual Information
10/2018–09/2019
Tokyo, Japan

Visiting Scholar with Prof. Rui Fan and Prof. Dan Alistarh
ShanghaiTech University (上海科技大学)
Large-Batch Algorithms for Deep Learning at Scale
04/2018–07/2018
Shanghai, China

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot
Technicolor AI Lab
Intelligent Visual Effects Scheduling via Constrained Matrix Factorization
09/2017–03/2018
Palo Alto, CA, USA

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller
Swiss National Supercomputing Centre (CSCS)
Distributed TensorFlow: A performance evaluation
06/2017–09/2017
Lugano, Switzerland

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308)
Teaching Assistant, UCPH: Advanced Topics in Natural Language Processing (NDAK19001U)
Teaching Assistant, UCPH: Neural Information Retrieval (NDAK20002U)
Spring 2017
Winter 2020
Spring 2021

Publications

8. **On Language Models for Creoles**
H. Lent, E. Bugliarello, M. de Lhoneux, C. Qiu, A. Søgaard
In *CoNLL 2021*
7. **Visually Grounded Reasoning across Languages and Cultures**
F. Liu*, E. Bugliarello*, EM. Ponti, S. Reddy, N. Collier, D. Elliott
In *EMNLP 2021* [Best paper award]
6. **Vision-and-Language or Vision-for-Language? On Cross-Modal Influence in Multimodal Transformers**
S. Frank*, E. Bugliarello*, D. Elliott
In *EMNLP 2021*
5. **Multimodal Pretraining Unmasked: A Meta-Analysis and a Unified Framework of Vision-and-Language BERTs**
E. Bugliarello, R. Cotterell, N. Okazaki, D. Elliott
TACL 2021
4. **The Role of Syntactic Planning in Compositional Image Captioning**
E. Bugliarello, D. Elliott
In *EACL 2021*
3. **It's Easier to Translate *out of* English than *into* it: Measuring Neural Translation Difficulty by Cross-Mutual Information**
E. Bugliarello, SJ. Mielke, A. Anastasopoulos, R. Cotterell, N. Okazaki
In *ACL 2020*
2. **Enhancing Machine Translation with Dependency-Aware Self-Attention**
E. Bugliarello, N. Okazaki
In *ACL 2020*
1. **Matrix Completion in the Unit Hypercube via Structured Matrix Factorization**
E. Bugliarello, S. Jain, V. Rakesh
In *IJCAI 2019*

Invited Talks

- **Visually Grounded Reasoning across Languages and Cultures** CIMEC, University of Trento 06/2021
Samsung AI Center, Toronto 11/2021

Skills

Programming

- Python, Matlab
- NumPy, Pandas, scikit-learn
- PyTorch
- CUDA, SQL

Languages

- native: Italian
- fluent: English (TOEFL iBT 109)
- intermediate: French
- basic: Chinese (HSK III), Japanese, Danish

Fellowships, Honors, and Awards

EuroHPC Grant with Desmond Elliott on “Vision and Language on a Shoestring without Tear”	06/2021
University of Copenhagen Marie Skłodowska-Curie TALENT Doctoral Fellowship	06/2019
EPFL PhD Program in Computer Science Fellowship [offer declined]	05/2019
The Japanese Government (MEXT) Scholarship [offer declined]	01/2019
PoliTong Merit Scholarship	07/2013
MIUR National Register of Excellences	06/2012

Service

Reviewer: EMNLP, ICLR, IJCAI, NeurIPS

Organiser: Multilingual Multimodal Learning Workshop (ACL 2022)